GE Medical Systems

Technical Publication

Direction 2352974-100
Revision 3

LightSpeed 4.X
Application Tips and Work-Arounds

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DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write “Damage In Shipment” on ALL copies of the freight or express bill BEFORE delivery is accepted or “signed for” by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (262) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage. Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins. 14 July 1993.

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment. Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

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LEGAL NOTES

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REVISON HISTORY

Revision	Date	Reason for change
0	9/02/2002	Pilot release 400.2_H16M3
1	9/10/2002	Updates based on M3 feedback 400.2_H16M3
2	12/06/2002	Full Production Release 401.8_H4.0M4
3	11/14/2003	404I.2_H4.0M5 + patch

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FEATURES VERSION 404I.2_H4.0M5 SOFTWARE

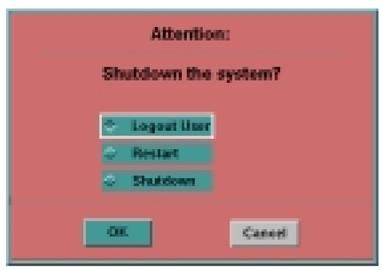
SOFTWARE VERSION

LS2002M5 – This release of software contains fixes to address a number of issues that were reported on the LS2002M4 release concerning reconstruction retires for Scout images, intermittent loss of alignment lights and prescribed tilt capabilities, Show Localizer not displaying or having a mismatch, ScanRx shutdowns, AutomA changes to improve reliability, image quality fixes for ring artifact with Bone and Bone Plus algorithm, improved availability of images in the AutoLink viewport, improved batch install for biopsy mode, selection of remote archive device, SmartPrep failed to start, IQ improvements for the 0.938:1 pitch and AutomA Noise Index set to 99 or Max mA set to mA.

There are some new features as well as enhancement to existing features.

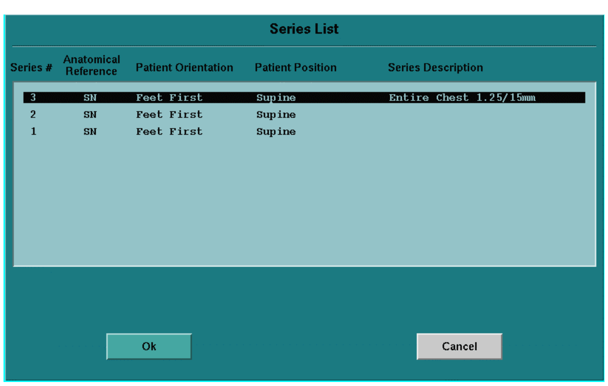
CHANGES TO SHUTDOWN

Shutdown - When shutdown is selected, you will be presented with 2 choices Restart or Shutdown. Restart shuts the system down and then automatically restarts the system. You no longer need to “Press any key to restart” the system. Shutdown brings the system down to the power off prompt. When shutdown is selected, you can to cycle power to the operator console to reboot the system.



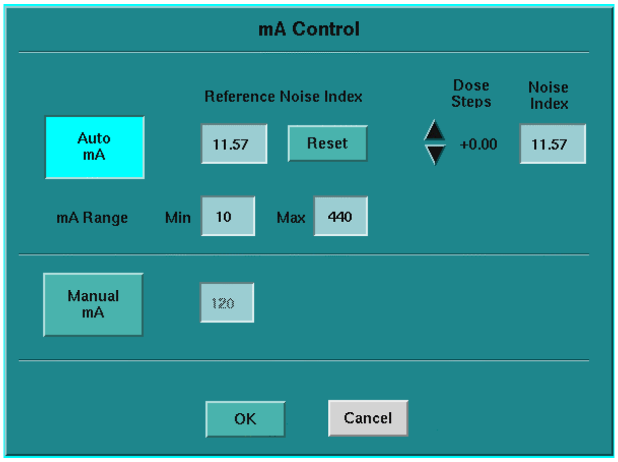
REPEAT SERIES

Repeat Series - Additional functionality for Repeat Series has been added for this release. Repeat Series now allows you to choose any series that has been scanned. When more than 1 series has been scanned a list of all scanned series will be displayed. Click on the series that you wish to repeat. If only one series has been scanned, Repeat Series will not display the Repeat Series menu.



AUTO MA

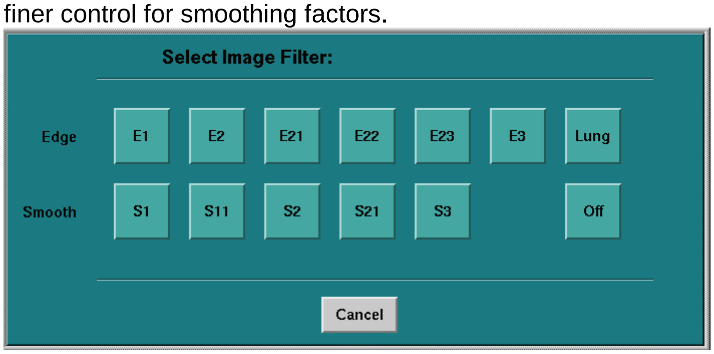
AutomA - changes have been made to the user interface to simplify the selection of parameters.



Reference Noise Index	Default or baseline Noise Index for the given protocol. Any changes to Dose Steps, Slice Thickness or Noise index are referenced to this value. This value can only be defined while in Protocol Management.
Noise Index	The noise level required for the study. As the Noise index increases the required mA decreases and image noise increases.
Dose Steps	Adjusts Noise Index by steps of 5%. Dose steps can be increased or decreased + values decrease image noise thus increasing required mA. Minus values increase Noise index, thus decreasing required mA. A Dose step value of 0 indicates that the prescribed Noise index is equal to the Reference Noise index for the protocol.
Reset	Resets the Reference Noise Index to the GE Target Noise Index Default for the anatomical area and slice thickness chosen in the protocol

PROVIEW

ProView Filters – the number of filters have been expanded. For Edge, the filters E21, E22, E23 have been added for finer control for edge enhancement factors. For Smooth, the filters S11, S21 have been added for finer control for smoothing factors.



LOW SIGNAL CORRECTION

Advanced Artifact Reduction – low signal reconstruction correction has been updated for this release. A reduction in streak artifacts through shoulders and hips should be seen.

DOSE PROFILE UPDATE

16X.0.625 detector configuration – Improvements have been made to the helical reconstruction algorithm for data acquired with 16X0.625 detector configuration. This change makes mA selection more logical when reconstructing the same slice thickness comparing 16X0.625 and 16x1.25 detector configurations.

The system will automatically update mA as slice thickness, pitch and table speed are changed. Auto mA has been updated as well.

With this change the mA required between detector configurations has changed from the previous release. Please see the following table for the differences. In most cases less mA is required than the previous release. In a select few cases more mA may be required.

Updates have also been made to the Channel Usage table to reflect the mm worth of data used to reconstruct each slice thickness.

mA factors:

Multiply the mA that you use today by the factor listed in the table below. This will update the technique used to account for the dose profile change.

5mm Full	mA Factor	Pitch	Table Speed
	.65	0.5625	5.625
	.8	0.9375	9.38
	.8	1.375	13.75
	.8	1.75	17.5
5 mm Plus			
	.85	0.5625	5.625
	.75	0.9375	9.38
	.75	1.375	13.75
	.8	1.75	17.5
3.75 mm Full			
	.8	0.5625	5.625
	.9	0.9375	9.38
	.9	1.375	13.75
	No Change	1.75	17.5
3.75 mm Plus			
	.65	0.5625	5.625
	.8	0.9375	9.38
	.7	1.375	13.75
	.9	1.75	17.5
2.5 mm Full			
	1.3	0.5625	5.625
	1.25	0.9375	9.38
	.8	1.375	13.75
	No Change	1.75	17.5

MA FACTORS CONT.

2.5 mm Plus	mA Factor	Pitch	Table Speed
	1.1	0.5625	5.625
	1.1	0.9375	9.38
	No Change	1.375	13.75
	1.10	1.75	17.5
1.25 mm Full			
	No change	0.5625	5.625
	2	0.9375	9.38
	.95	1.375	13.75
	No Change	1.75	17.5
1.25 mm Plus			
	.9	0.5625	5.625
	1.4	0.9375	9.38
	.9	1.375	13.75
	No Change	1.75	17.5

CHANNEL USAGE TABLE FULL MODES:

Full Modes									
Scan Mode		Image Thickness							
Helical Pitch	Table Speed (mm/rot)	Detector row collimation (mm)	0.63mm	1.25mm	2.5mm	3.75mm	5.0mm	7.5mm	10.0mm
0.563:1	5.625	0.625	0.62	2.18	4.06	5.92	7.85		
	11.25	1.25		1.62	4.48	6.35	8.20	12.08	15.70
0.938:1	9.375	0.625	0.93	2.27	4.24	6.14	8.17		
	18.75	1.25		1.87	4.73	6.73	8.59	12.27	16.33
1.375:1	13.75	0.625	1.24	2.57	4.53	6.47	8.48		
	27.5	1.25		2.49	5.34	7.32	9.36	13.16	16.96
1.750:1	17.5	0.625	1.24	2.58	4.54	6.44	8.47		
	35	1.25		2.49	5.34	7.33	9.21	12.88	16.93
0.625:1	6.25	1.25		1.24	4.45	6.80	5.74	8.10	10.60
	12.5	2.5			2.49	6.78	11.03	15.70	16.72
0.875:1	8.75	1.25		1.24	4.38	6.94	5.74	8.11	10.61
	17.5	2.5			2.49	6.79	11.04	15.70	16.72
1.350:1	13.5	1.25		2.50	6.24	7.18	7.23	9.60	11.98
	27	2.5			4.99	9.47	11.69	14.98	14.02
1.675:1	16.75	1.25		2.49	4.97	6.33	7.81	9.57	11.92
	33.5	2.5			4.97	9.60	9.43	14.64	13.96

CHANNEL USAGE TABLE PLUS MODES:

Plus Modes									
Scan Mode		Image Thickness							
Helical Pitch	Table Speed (mm/rot)	Detector row collimation (mm)	0.63mm	1.25mm	2.5mm	3.75mm	5.0mm	7.5mm	10.0mm
0.563:1	5.625	0.625	1.30	2.66	4.76	7.00	9.28		
	11.25	1.25		2.68	5.22	7.48	9.65	14.20	18.57
0.938:1	9.375	0.625	1.65	2.73	4.98	7.36	9.48		
	18.75	1.25		3.41	5.60	7.91	10.10	14.71	19.46
1.375:1	13.75	0.625	1.98	3.07	5.34	7.64	9.66		
	27.5	1.25		3.97	6.20	8.47	10.73	15.34	19.31
1.750:1	17.5	0.625	1.96	3.04	5.28	7.65	9.76		
	35	1.25		4.02	6.22	8.50	10.71	15.29	20.03
0.625:1	6.25	1.25		2.24	6.17	8.24	10.23	10.79	12.72
	12.5	2.5			4.67	8.35	11.47	15.46	18.09
0.875:1	8.75	1.25		2.36	4.72	8.47	10.24	10.80	12.73
	17.5	2.5			4.84	8.48	11.22	15.95	17.73
1.350:1	13.5	1.25		3.87	6.45	9.25	10.61	12.98	13.86
	27	2.5			7.35	9.99	13.47	15.83	16.60
1.675:1	16.75	1.25		3.97	6.42	9.29	10.67	13.04	13.91
	33.5	2.5			7.93	10.18	13.28	15.52	16.75

AUTO TRANSFER

Auto Transfer - by image has been enhanced to transfer images in larger groups. This provides more efficient use of system resources while Auto Transferring by image. Auto transfer by image will transfer the first 50 images in 2 groups of 25, after that images will be transferred in groups of 50.

ANONYMOUS SCAN DATA SAVE

Anonymous Scan Data Save - The system now provides the ability to save scan data with patient identifying information anonymized. This has been added to address HIPAA concerns about patient confidentiality. Scan files (raw scan data) saved anonymously and then restored will reconstruct with patient identifying information removed. The Patient Name and ID will be anonymized.

Select Recon Management

Select Save Scan Data

Select Save Selected Anno. Data to save scan files as anonymous data.

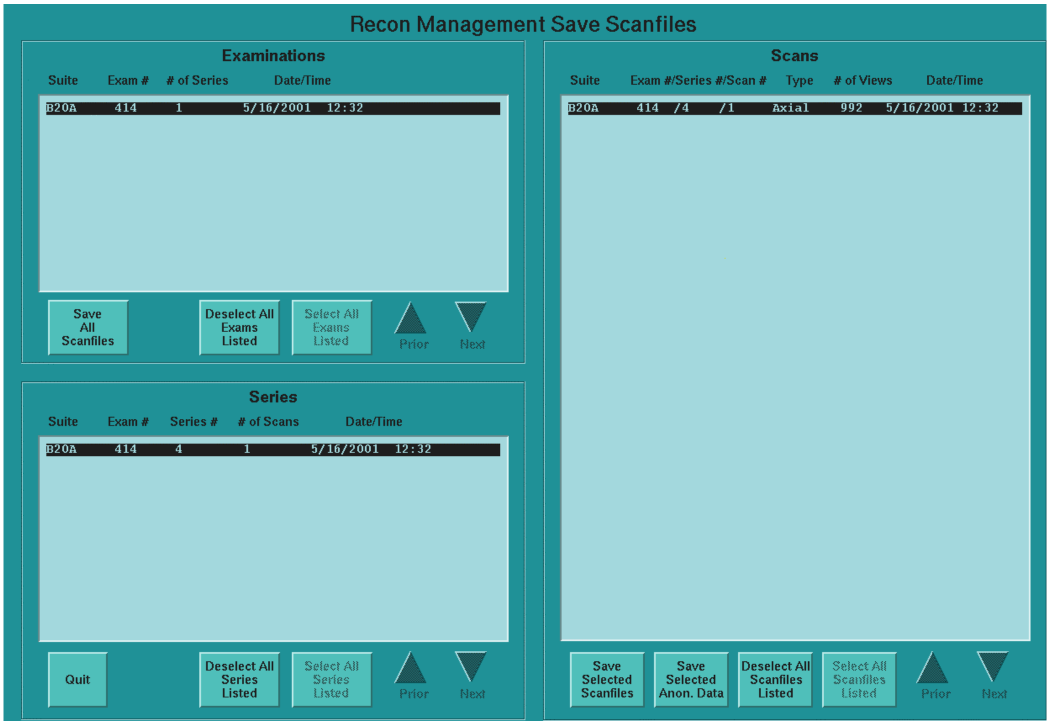
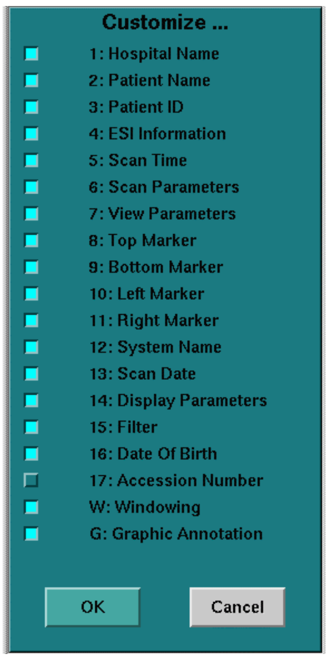


IMAGE ANNOTATION

Accession Number – Image annotation field Req No.: has been replaced with annotation which more appropriately identifies it as the field for the accession number entered on the New Patient screen. The annotation on the image is Acc No.:.

CUSTOM ANNOTATION

Custom Annotation – has added an additional custom annotation field for Accession Number. Selecting group 17 will allow the user to enable or disable display of the Accession Number field on the monitor or on film.



NEW PRECAUTIONS 404I.2_H4.0M5

SYSTEM

- The scan hardware may fail to reset on reboot. Select Service Desktop, System resets, Scan, Run to reset the scan hardware. If the reset still fails, then shutdown and reboot the system to recover.
- The system may not show typed information on the New Patient or ViewEdit screen. The system will update after a short time.
- The system may be slow to respond to keyboard entry and mouse clicks. If the system fails to respond stop and let the events catch up.
- The first character in Exam description may not be shown in the text field but is shown in the Browser description.

SCAN

- Show localizer does not show the Scan Field of View when prescribing scans. When scanning head studies, verify that you have your patient centered at ISO center to assure that the entire head is included in the SFOV.
- Show Localizer may fail to display. If toggling the Show Localizer button does not display the scout, then display the scout in a free viewport and use Report Cursor to find your scan locations and A-P and R-L centers, manually enter these on the Viewedit screen.
- Show Localizer may remove the localizer lines when page down or page up is selected and will not advance to the next scout. To recover from this, toggle Show Localizer off /on. If Scout Image 2 is displayed use Page Up, to go to scout image 1 use Page Down.
- Show Localizer may display a scout from a previous patient along with the current exam scout. Even though this scout is displayed, the scan prescription lines will not be shown on this scout.
- The Show Localizer button may get in a stuck state and not display a scout image if the last scout scanned has an azimuth other than 0,90, 180, 270. Manually display a scout in a free viewport and user report cursor to find your scan locations and then manually type in the Start and End location desired.
- A scan abort may occur between 2 scan groups with gating enabled. To avoid this prescribe the shortest delay possible.
- Auto Voice may fail to play for prescriptions with 2 groups and Auto mA enabled or with manual mA when the mA changes between the groups. An Auto Voice has failed to play message will be posted on the right monitor is the message area on the upper left of the right screen.
- The pause button on the screen in Tube warmup does not function. Use the stop scan button on the SCIM.
- You may be prompted to run DAS gain cal prior to starting Fast Cal.
- The system may not allow changing of the landmark even though no scans have been taken in the series. A landmark that is set at the edge of the scannable range most often causes this. To reset the landmark, choose Select New Protocol.
- Always verify the mA table prior to scanning when using Auto mA.
- The dynaplan screen may fail to update if Priority Recon is selected during scanning

Scan continued

- Auto mA may set the min and the max value to 10 when selecting a protocol or adding a group. Always verify the mA tables prior to confirming the scan to verify the mA values in the table are reasonable for the patient being scanned. If the values do not look reasonable select the mA button and verify the Auto mA parameters.
- Auto mA may show the max mA from a previous group instead of the max mA prescribed. Always verify that the max mA parameters are correct before confirming a scan.
- Auto mA may set the Max mA to 50. Always verify the mA table prior to scanning.
- The gating button may turn red when the patient experiences a big jump in heart rate such as in a PVC. The system usually will recover and read the signal. Sometimes you may have to toggle the gating button Off/On to get the heart rate signal synched again. Care should be taken in starting the scan if the patient continues to have PVC's
- The cut lines in Show Localizer and in Cross Reference in the vertical presentation appear multi-colored on the monitor. The lines are a solid color when printed on film or screen saved.
- Selection to move group of slices may be sluggish is GraphicRx.
- Heart rates of greater than 120 are not valid for prospective gating for SmartScore acquisitions. The system will not stop scanning if the heart rate exceeds 120 BPM; however if the user observers higher than acceptable heart rates, they should pause the scan and resume when heart rate is in a valid range.
- The Start/End locations and Gantry Tilt fields may show strange characters when |Add Group| is selected quickly after |Show Localizer| is selected and a Direct Vis application (Direct 3D, VariViewer) is enabled for the series. Wait until the localizer is displayed before performing any edits for the series.
- Scan Abort may occur if firmware detects the Axial drive not ready. Select Resume and continue.
- An artifact resembling a tire track may be seen using the 0.562:1 pitch with the 16x0.625 detector configuration making 5mm slices using a 1 second rotation speed with 120kVp and 330mA. Consider changing rotation speed or detector configuration if you have protocol that uses these parameters to avoid the artifact.
- In Recon2, the system may fail to show the last slice prescribe in Recon1.
- AutoVoice may fail to play if the table is moved between groups where a long groupdelay has been prescribed.
- Alignment lights may fail to turn on. Perform a system reset. If alignments lights still fail to turn on. Shut the system down and turn off power at the main breaker and then restart system.
- Show Localizer may remove the slice lines when page down or page up is selected and may not advance to the next scout image. To recover, toggle Show Localizer off / on.
- System performance may become slow if the optimize screen for mA is left open. Make selection in the optimize pop-up screen and then close the window by clicking on Optimize button.
- Selection of up front delay in Optimize is very slow, it may take 10-20 seconds for the system to respond.
- Add group may not work. Try selecting One More and changing the number of slices to desired number.
- In GraphicRx, it maybe difficult to select a group that is inside the range of another group. Select group on the ViewEdit screen to make that group active in Show Localizer.
- If you have two protocols within the same exam that have different patient positions, the system may not move to the correct start location for the series with changed patient position. If patient position is going be changed in regards to head first verses feet first, please start a new exam.

TUBE WARM UP/FAST CAL

- Fast Cal may give error that Collimator Cal failed. Resume Fast Cal and contact GE Service.

PERFORMED PROCEDURE STEP (PART OF CONNECT PRO OPTION)

- The PPS server may stop and fail to send the request. You may have to select Complete multiple times to get the series to update to the complete state.
- Add Sub does not update images for PPS correctly
- Perfusion2 will not launch if images contain PPS information.
- If the total number of prospective images in an exam is greater than 2500, PPS may fail to complete.

PROTOCOL MANAGEMENT

- The Dose information in protocols used from Most Recent does not reflect the dose actually used for the protocol. If the protocol contains manual mA, retype the mA value to update the dose display. If Auto mA is used, the dose will not be reflected in Protocol Management, but will be updated at scan time for the patient being scanned based off a valid scout image.
- Protocols can be restored from previous LightSpeed Ultra, LightSpeed Plus or LightSpeed QX/i to LightSpeed^{16} systems however any 4-slice protocols will be converted to 16 slice modes. The following conversion will be used:
  - Thin Twin Helical 16 row 0.5625:1 pitch, 5.625 table speed, 0.625 thickness Full mode.
  - 0.75:1 HQ, 8 row 0.625:1 pitch, 6.25 table speed, for 4x1.25 and 4x2.5 detector configuration Full mode
  - 0.75:1 HQ, 8 row 1.35:1 pitch, 13.5 table speed for 4x3.75 and 4x5 detector configuration Full mode
  - 1.5:1 HS, 8 row 1.35:1 pitch, 13.5 table speed for 4x1.25 and 4.25 detector configuration Full mode.
  - 8 row 1.35:1 pitch, 27 table speed for 4x3.75 and 4x5 detector configuration Full mode

SMARTPREP (PURCHASED OPTION)

- SmartPrep may fail to confirm to take the baseline image. Try confirming again, if this does not clear the problem then a system reboot will be needed to get SmartPrep working again.
- The SmartPrep baseline image may fail to display even though it is reconned and in the database. Go back to viewedit and rescan the baseline image to continue.The Monitor phase image and graph information will not update if the table height has changed from the Baseline to Monitor phase. Do not change the table height after the baseline scan has been taken to avoid this.
- When transitioning to the Scan Phase from Monitor phase if the XR Auto View layout is selected the upper left viewport will be black. Move your mouse cursor over the viewport to redisplay the image.

RECONSTRUCTION

- Streak artifacts may be seen in images when a tube spit has occurred.
- Image recon may fail and then shutdown. Select Recon Management, select Pause queue, Restart queue, if this does not restart recon then a Restart of the system will be needed.
- Cardiac recon modes of Burst 2 or Burst 4 cannot be selected via column edit. Please select the individual row to edit recon mode selection.
- The dynaplan screen may fail to update if Priority Recon is selected during scanning
- Anatomy scanned off center from ISO center such as extremities, may have loss of resolution when data is acquired helically. Always make sure to center to anatomy of interest as close to ISO center as possible.
- Cine acquisitions may have retries or image suspension. Unsuspend images in recon management.

RECON MANAGEMENT

- Scan File save is not complete even though the Saved File message has been posted. The problem is due to the time it is taking to unmount the MOD. You may have to wait 5-7 minutes for the MOD to unmount. The write light on the MOD will be green when the MOD is unmounted
- Scan Files restored from a different system will be reconstructed using the product name of the system it is restored on. For example, restore scan files from a H1 SDAS QX/i or a H2 SDAS Plus to a LightSpeed[16] and the product name on the images will be indicated as LightSpeed[16] instead of LightSpeed QX/I or LightSpeed Plus. All other patient and parameter information is correct.

DISPLAY (APPLIES TO BOTH IMAGE WORKS AND EXAM RX DISPLAY)

- Plus recon mode is not annotated on the Series Text Page. Refer to the image annotation for the Plus recon mode annotation.
- An active user annotation graphic will not film the box or arrow that is displayed on the screen.
- Font size on the ROI Text Page is smaller compared to prior release of software.

IMAGE WORKS DISPLAY

- The primary focus may change if the space bar is selected multiple times when entering accelerator line commands. Verify that primary focus is on the image desired.
- The Accelerator Line is a dark green in Image Works instead of a light green as on the ExamRx desktop. It is active and will accept commands.
- When changing from 3D to Navigator, sometimes the threshold value is automatically set by the system and the user is unable to change it.

EXAM RX DISPLAY

- The Trackball may fail to initialize properly at system reboot. Try rebooting the system again to recover
- The Icon for paging/WL and Auto Link is not visible
- Display windows may display on the left screen instead of the right. This will most often occur when switching desktops rapidly. A system reboot will be needed to recover.
- Pressing page up 2 times causes the ww and wl to change on the scout on it's own. Readjust the ww and wl to continue.
- The first time the trackball is used for paging after a system reboot window level will be changed instead of paging the images even though the P icon is displayed. Click once to switch to W mode then click again to place the trackball in paging mode.
- The AutoView or Auto Film viewport may display a very magnified image. This image may be filmed in Auto Film. Review your films and verify that all filmed images look correct.
- Viewport format may not be able to be changed and requires switching layouts to correct. This occurs if F3 or Shift F3 is used for filming and next image is used. The viewport format is now locked. You will have to switch to another review or Auto View layout to correct.
- MIROI may fail to produce a graph if next prior is selected while creating the MIROI. Redisplay the series and begin again.
- Cross Reference image may not print sometimes instead the last image will be printed twice.

VARIVIEWER

- Batch film may fail after a system reboot if the format selected is different than the default film format on reboot. An error dialog will be posted informing you of the error. You may have to select filming multiple times to successfully send the request.
- If the scout is taken after the series used for VariViewer operation, the system may be unable to find the scout for graphic batch prescription. Manually enter the range for the batch prescription.
- Any zoom applied to a Variviewer image is not maintained as you page through the images.
- VariViewer session may not be able to be started if there multiple Pause scans during the acquisition of data.
- VariViewer is not compatible with a gantry tilt.
- Switching back from a 768 viewport is VariViewer causes the VariViewer viewport to be forward instead of the Auto View viewport. Use the page turner to toggle back to the auto view Viewport.

DIRECT3D - OPTION

- Batch film may fail after a system reboot if the format selected is different than the default film format on reboot. An error dialog will be posted informing you of the error. You may have to select filming multiple times to successfully send the request.
- 768 image size may display split in 2 if resize is selected with the Auto View viewport forward. Always have the D3D viewport forward when selecting resize. To recover from this problem select resize twice.

Direct3D continued

- If you resize the viewport in Direct3D, the image may not initially be displayed. User the Page Turner in the lower left corner of the viewport to toggle to the time.
- The Save State, which allows the user to rebuild the Direct3D model on the AW using the VR algorithm from D3D, does not work on the OC even if Volume Viewer is installed. The user will get an error message informing that the images are missing.

REFORMAT

- Reformat will only allow filming in Batch to the first 30 cameras in the list. If another camera is desired save the batch images to disk and use Print Series to film the images.
- Images saved from batch mode may appear to have jagged edges or not have optimal Image Quality. Manually save the images to improve image quality.
- Large image series may fail to load when using Reformat detail. This can be caused by small spacing interval between images or image sets greater than 1000 images. Select a sub set of images if this occurs or use Reformat Standard to build the model.
- A series that has more than 1 group with 0.625mm slice thickness will fail to load for reformat or will fail in Volume rendering on the AW. This is due to a mismatch in the spacing between the groups. To avoid this have only one scan group if possible, otherwise selectively highlight the images from the first group for Volume Analysis.
- Sagittal images created from a head first prone data set scanned in the direct coronal position will appear elongated.

CT PERFUSION 2 ON OC – PURCHASED OPTION

- Computation of perfusion maps and automatic detection of the artery or vein may not function when the data displayed in Perfusion is initially black and the WW and WL values need to be set to large negative values (WW 40, WL –1020) to make the images viewable. The threshold should be set to the maximum negative value for air to assure that automatic detection of the artery and vein and that Perfusion maps are displayed correctly.
- Perfusion will not launch if the images contain PPS information
- Saving TIFF files of the graph and images is not supported because there is no floppy drive hardware available on the OC.
- Set the color ramp to gray scale for screen saves images if you are going to network the images to another station such as PACS to improve quality of the screen save.
- Initially as CT Perfsuion2 is loading, you may see garbage flash along the left side of the screen.
- Processed data does not display if New Protocol is selected within Perfusion2. To avoid this always close Perfusion2 before selecting a new protocol.

ADD SUBTRACT

- Comb images will not contain the e/s/i used for the added image.

ARCHIVE

- Even though the Sony MOD drive states that it is a 5.3 gb drive. The archive software only supports a maximum size of 2.3 gb. 2.3 gb 512 bytes per sector is the preferred archive media size to utilize on the system.
- Archive may fail and slow the system when more than 200 images are queued by image. Try to archive by series if possible to avoid this.
- The system may report that the media is full even though the media has just been labeled when Save by image is selected. Try selecting a smaller range of images to avoid this.

FILMING

- Image setting such as WW WL and flip rotate, zoom, roam are not maintained across all images when imitating F4 print series from a MID Viewport format. Use a 512 size viewport to initiate F4 Print Series.
- Auto Film may stop and the Continue Same Sheet button will be displayed. The Continue Same Sheet button will be active, but when selected will not restart filming. Use Manual Film to complete filming the exam and reboot the system before starting the next exam
- If the F1 key is used to film the Text Page ROI, Exam Text Page and Series Text Page, the text pages will be filmed to the film composer last selected in the pop-up for text page. Use the filming selections in the text page screens to film the information displayed.
- The message “Film formatting in progress. Please retry.” May be seen when filming to the Manual Film composer when the system is busy with other simultaneous operations such as recon, network, auto film and scanning. Retry manual filming when the system is less busy. A system reboot may be required to recover.
- Image annotation may be small or blurry on filmed images when the first image is a scout that is over 600mm. Place this image on the last space of a film to avoid this.

NETWORK

- The system may create core files that may fill up disk space on the system. Networking color save images from Volume Viewer or CardiacIQ causes these core files. To avoid this save the color images to MOD, remove the color images from the system then restore the color images and then network the images.
- Color Saved images from Volume Viewer or CardiacIQ may fail to transfer. To avoid this save the color images to MOD, remove the color images from the system then restore the color images and then network the images.
- Images that have been edited using Edit Patient and then networked to a RadStore system will fail to archive to the RadStore DVD. The images must be archived on the CT system MOD.
- The system no longer supports Advantage Net protocol. When sending images, DICOM send should be utilized.

IMAGE MANAGEMENT

- Removal of images should be done with the system in an idle state. Failing to remove images at idle times can cause image install timeouts that can then cause recon to shutdown and can cause the browser to fail to respond. The user may also see a blank list in the browser. The system should not be shutdown until the image space has been updated fully from the remove. The length of time to synchronize the database is proportional to the number of images being removed. This will assure that all images are reconstructed, displayed and installed into the database without error and that image space is update correctly.

LEARNING AND REFERENCE GUIDE

- The Learning Solutions CD-Rom may not launch the first time. Wait a minute and try again.
- After a desktop switch back to Learning Solutions desktop the guide returns to the home page instead of returning to the page they were on when the switch occurred. The document is slightly behind the home page. Left click and drag on the home page border to move the home page so you may access the Learning and Reference Guide pdf file.

LEGACY PRECAUTIONS

SYSTEM

- Run Shutdown and Restart daily.
- The system may fail to start if system disk space needs to be recovered from unused system files. The following message will be posted in the Disk Management shell window:

  ```
  "The purpose of storelog is to recover system disk space by archiving and then
  removing core, log, and data files that have been saved for their system
  troubleshooting diagnostic value. Now they may be taking up too much space for the
  system to run properly. Removing these "system log" files does not add image space,
  but should allow the application to start."
  ```

- You are then directed to answer the following questions (be patient for the messages to display):

Place the mouse cursor inside the black Disk Management window

Q: Do you want to save system log files to removable media?

Type n and hit the enter key

Q: Do you want to remove the system log files?

Type Y and hit the enter key.

You will then see a message "`Please be patient, the system is now removing unwanted files`." The system will then start normally.

- If the system fails to startup completely, select [Unix Shell] from the tool chest menu in the upper right of the Scan Monitor place your cursor in the blue shell and type st.
- If mouse selections fail, select the [Escape] key to clear backlogged requests.
- If you can no longer type in a field, move the cursor to the field and try clicking the middle mouse to restore functionality. If this does not work, then a system shutdown will be required.
- The cursor may not move between the left and right screen. Wait a few seconds without moving the cursor and try again.
- To install a SMPTE pattern, BRH or Quality assurance images for viewing, select the following. Once installed the images will be available for viewing from the Image Works browser or List Select in Exam RX. Both will be listed as Exam 1000, the patient name will reflect if it is a QA image or a SMPTE pattern.

  Select Service Desktop

  Select [Diagnostics]

  Select [Display Processing]

  Select [Install SMPTE Image]

- In general, wait for a screen transition to take place before making another selection.
- If the console becomes unresponsive for 2 minutes or more, shutdown the system using the pink shutdown button and restart the system. If you cannot select the pink shutdown button, turn off the console power switch, wait 10 seconds, then turn the console power switch back on. The system should come up normally.

System continued

- The system will lock up if the mouse becomes hung on the left monitor. The only known recovery is to cycle power to the console.
- You may see a pink warning dialog posted indicating the system has low disk space. This is due to a partition on the system disk getting too full. This can be caused by archive, network or film queues containing too many entries. Make sure that the entries are processed. If the queue is paused, resume it or delete entries as needed.
- Protocol Management may not be selectable. Check that retro recon is selectable. If it is not then the system will need to be rebooted.
- Pop-up screens and menus may appear on the wrong monitor or may be displayed split between the two monitors.
- Windows such as the Film Composer that normally are restrained to the right or left monitor can be dragged to the other monitor. Be careful not to get a Film Composer hidden behind some screen if you move it to the other monitor.
- If a blank viewport is seen where image annotation is displayed and W/L is interactive but no image is seen, shutdown the system, recycle power and reboot the system to clear this display issue.
- If corrupted images are seen after applying an image filter such as enhance, smooth, lung or gray scale, try rebooting the system and see if the image can be displayed correctly. If not contact your GE service representative.
- The dynaplan screen may fail to update if Priority Recon is selected during scanning
- Pressing the Space Bar after selecting all fields in New Patient, locks the patient information fields and all the patient information fields become insensitive. Only the delete and backspace key will be active. Press the delete key to recover from this situation.
- Entering a \ and pressing the Enter key in the New Patient can cause the fields to become insensitive.
- Under heavy system operation, the message “Unable to install image in the data base” may be seen. Recon for the series will take longer due to unsuccessful install of images to the data base.
- Double and triple click for selection of primary or secondary focus may be slow. Wait for the system to respond before repeating the focus selection mouse commands.
- Auto Voice volume setting changes are not remembered across system reboots unless the change has been saved. The save a volume setting do the following:
    1. Select Image Works
    2. Select Auto Voice Volume
    3. Adjust the settings as desired
    4. Select File menu
    5. Select Save
    6. Select OK

TUBE WARM UP/FAST CAL

- Run Tube Warm up after 2 hours of non-use. It is best to warm the tube as close as possible to the time the next patient will be scanned. Don’t warm the x-ray tube and then wait 30 minutes without scanning.

Tube Warm Up/Fast Cal continued

- Always check that the beam is clear when doing Fast Cal. In Fast Cal, the system checks cleanliness of the Mylar Window. If the system suspects the Mylar Window may be dirty to the point that it may cause a beam obstruction, a pop-up message will be displayed. The operator will be instructed to clean the Mylar Window and select [Retry].
- Fast Cal must be run once every 24 hours.
- Complete all portions of Fast Cal, Warm Up I, Warm Up 2, Gen Cal, Clever Gain and Fast Cal. This assures that the Air calibration and generator calibrations are up to date on the system. Scan aborts may occur during Axial or Helical scanning. Always be aware of the scan progress during an Exam and select Resume as soon as it is posted to continue.

SCAN

Scan aborts may occur during Axial or Helical scanning. Always be aware of the scan progress during an Exam and select Resume as soon as it is posted to continue.

- Scan may fail to confirm posting a message that not enough image space exists, even though the image space shown in the Feature Status Area indicates there is enough space. This is due to the fact that images are stored on the system disk in more than one partition. Remove consecutive exams to free up image space for confirm to proceed.
- To terminate an Insite connection on your system. Select [New Patient] a message will be posted informing you that the scan hardware resource is not available. Wait 3 minutes and select [New Patient] again to begin scanning.
- In general, if a scan fails and a [Resume] is posted, press [Resume] to continue. Try [Resume] again if the first [Resume] fails. If a failure still occurs, reset the scanning hardware through System Resets in the Service desktop. If scan still fails to restart, shutdown and Restart the system.
- Auto Voice may fail to function, especially during system simultaneity. Make sure that you can hear the Auto Voice to recognize if Auto Voice has quit, manually breathe the patient when this occurs.
- The Show Localizer scout image may fail to display or images may fail to recon and may become suspended if scanning is started while remove images is in progress. To avoid this **don't** remove images while an exam is in progress or scanning is active. Remove images when the system is idle.
- The Dynaplan screen may report incorrect status (i.e. a scan is removed before it is actually scanned) when stop scan is selected. The screen will reset correctly after Resume or back to view edit is selected.
- Scan groups will not be contiguous if you switch from 2 or more Helical scan groups column or row edit and the scan type is changed to Axial. If this change is made, then verify the Start and End locations of each of your scan groups and adjust if needed.
- The Cine time between images may change when other parameters are modified within a protocol. Check the Cine time between images prior to confirming the scan to verify it is the value you wish.
- Tilt handles will not be visible on the screen if the DFOV is larger than 48. Use a DFOV smaller than 48 to assure that the tilt handles are visible.
- Show Localizer may fail to display the scout image if the Next Series is selected before the scout images are reconstructed. Wait until the scout images are reconstructed before selecting Next Series. If Next Series has already been selected, toggle Show Localizer off and then on again.

Scan cont.

- Quickly changing desktops when Show Localizer has been selected for the next series can cause a partial display of the scout in the Graphic Rx window to occur.
- If Page Up or Page Down fails to change the display for Show Localize:

  Try the following in order:

  1. Turn Show Localizer button off then on
  2. Move the cursor off the image window and back on to the image to refresh the screen.

  If both of these fail, then verify that you have a valid scout for Show Localizer.
- If an R, L or A, P value is more than half of the Display Field of View in mm. The image annotation will not annotate R and L or A and P on the images. For example if the DFOV is 10cm and the R value is 56mm then the image annotation will show R R. Use a R-L or A-P value which is less than half of the DFOV in mm to avoid this.
- A protocol may fail to display. The following message will be posted to the message line on the lower center of the scan monitor "Can't read selected protocol, please choose another protocol". If this occurs the protocol is corrupt and will need to be rebuilt. Delete the protocol and rebuild it.
- The patient history field may be missing the last 9 characters when all 60 characters have been entered and the patient record is either selected as a completed exam or the Patient ID is entered and the system matches a completed exam.
- Even weights entered in Patient Schedule may be rounded up to the next odd pound. This occurs because weights are stored as kilogram units then converted back to pounds.
- During the acquisition of large data sets (1000-1500), there may be Auto film and reconstruction performance issues. If either stops, check the queues and restart to continue.
- When selecting the Auto Voice field, the selected Auto Voice will be deselected. You must reselect the desired Auto Voice message before exiting the Auto Voice screen.
- When entering birth dates on the New Patient or Schedule Patient screens the following rules apply:

  - 2 digit years can only be entered if the birth year is in the current century for the 21st century **-** for the year 2000 forward.

  - Birth dates from any other century must be entered as 4 digits for example for the year 1899 –1999 all four digits must be entered.

  - Birth dates can only be entered for 150 years (Current year minus 150 years).
- If InSite is running the "Remote Safety" test, New Patient will not open. New Patient can be made available by either:

  1. Calling InSite and request they abort the test.
  2. Go the the Service Desktop and select |CleanUp| option to cancel the test in progress.
- PMR images may fail to recon id disk space is low. To recon the images, remove images and restart the recon queue in Recon Management.
- You may get the message "Duplicate Scan Key", if a Scout Scan is aborted within the first 20 to 75 mm of table travel or if a the prescribed length of the Scout is less than 75mm. In order to proceed, an End Exam must be done and new exam started.
- It may be possible to prescribe a recon range that is outside the Recon 1 boundary. This may occur if the start location is dragged below the recon end location in Recon 2 or 3.

Scan continued

- If a large number of scan files are reserved and there is a large recon queue, you may see the message: “Duplicate Scan Key – unable to allocate scan file.” There will be a resume at Start Scan. Either wait a few minutes for the recon queue to decrease or release some of the reserved scan files.
- Bands and lines may be seen in scout images if there is a tube spit or the reference channel is blocked during the acquisition of a scout scan.
- A zero interval axial series, if paused, will add another group equal to the remaining number of images. Make sure to return to the View/Edit screen to delete this group to avoid acquiring additional images.
- Biopsy is not valid for a Thin Twin helical acquisition mode. If Biopsy is selected, the Biopsy window does not open; however a group is added to the series at an inadvertent location. Do not select Biopsy mode with Thin Twin helical scan mode.
- If the message “Can not read Cal Database”, select |End Exam| and re-enter patient information.
- A scan abort may occur, resume to contiune. These can be caused by the following

  -Waiting more than 103 seconds to press the move to scan button after confirm. To avoid this do not confirm until everything including the patient is ready for the scan and when prompted move the table into position.

  -Taxi link violation due to forward error correction

  - DIP waiting to receive buffer of views
- If the scout is not displayed but available in the Browser and you get the message “Patient Position has changed” or “No matching Scout found” for show localizer. The system will not be able to perform graphic prescription. Use explict prescription to complete the prescription or end the exam and start a new exam.
- If Direct3D or VariViewer are enabled in the DirectVIS under the Recon tab, Add Group will display some fields as insensitive as it will be combined with the current Direct3D or VariViewer session. Turn Direct3D or VariViewer off for the added group if you no longer what it to be included as part of the Direct3D or VariViewer session. This will allow changes to any of the acquisition parameters.
- Auto mA should not be used with Gating and Cine or Cardiac Helical acquisitions even though it can be selected.
- The gating button may turn red when the patient experiences a big jump in heart rate such as in a PVC. The system usually will recover and read the signal. Sometimes you may have to toggle the gating button Off/On to get the heart rate signal sych’d again. Care should be taken in starting the scan if the patient continues to have PVC’s.
- Hitting the F12 key while in Show Localizer can cause ScanRx to crash. Once the system is in the Show Localizer mode to perform graphic prescription, the F12 key has no functionality. F12 key is valid only in AutoReview and Review Layout formats.

PROTOCOL MANAGEMENT

- Protocols which contained a SmartStep series cannot be selected or copied from the Most Recent selector.

Protocol Management continued

- Protocols can be restored from previous LightSpeed Ultra, LightSpeed Plus or LightSpeed QX/i; however any 4 slice protocols will be converted to 16 slice modes. The following conversion will be used:
  - Thin Twin Helical 16 row 0.5625:1 pitch, 5.625 table speed, 0.625 thickness Full mode.
  - 0.75:1 HQ 8 row 0.625:1 pitch, 6.25 tablespeed, for 4x1.25 and 4x2.5 detector configuration Full mode
  - 0.75:1 HQ 8 row 1.35:1 pitch, 13.5 table speed for 4x3.75 and 4x5 detector configuration Full mode
  - 1.5:1 HS 8 row 1.35:1 pitch, 13.5 table speed for 4x1.25 and 4x2.5 detector configuration Full mode
  - 8 row 1.35:1 pitch, 27 table speed for 4x3.75 and 4x5 detector configuration Full mode
- Show Localizer On is not saved with protocols in the Most Recent list even though it was turned on for the protocol. Edit the protocol after it is selected or copied to turn Show Localizer on.
- There is no default protocol for Pediatric areas Neck, Upper Extremity, Chest, Abdomen, Spine, Pelvis, Lower Extremity.
- If the weight of a child is at the cross over point of a weight category, due to rounding errors the correct weight category might not be selected. Please check the weight based category selected against the label and the patient's weight.

SMARTPREP (PURCHASED OPTION)

- Monitor phase scanning will abort if the Monitor ISD is set to less than 3 seconds when the rotation time used is 0.8 sec. Set the Monitor ISD to 3 seconds or greater to avoid this problem.
- If this has occurred, do one of the following:
- Press Monitor Phase to continue monitor scans.
- Use the graphed data and the clock countdown on the SmartPrep display to determine when to select Scan Phase.
- The film composers will flash and cannot be hidden on the SmartPrep display. This will occur if the Auto Film or Manual Film composer buttons are selected after confirm has been selected for SmartPrep. Don't select the film composer buttons while the SmartPrep display is first coming up or is visible to avoid this.
- If reconstruction fails during SmartPrep, scanning will pause and post a message. Press |Monitor Phase| to resume scanning for the monitor phase.
- SmartPrep may fail to confirm for the Baseline scan. Re –confirm to continue.
- The SmartPrep baseline image may fail to display even though it is reconned and in the database. Go back to viewedit and rescan the basline image to continue.
- The Monitor phase image and graph information will not update if the table height has changed from the Baseline to Monitor phase. Do not change the table height after the basline scan has been taken to avoid this.
- When transitioning to the Scan Phase from Monitor phase if the XR Auto View layout is selected the upper left viewport will be black. Move your mouse cursotr over the viewport to redisplay the image.

PATIENT SCHEDULE

- Close the preferences screen before switching between Patient Schedule and New Patient.
- The Patient Name, Patient ID, Accession Number and Requested Procedure ID can only be edited if “Allow to Edit MWL” is set to yes in the Preference window.
- The Patient Schedule button may not display the Work List from the HIS/RIS server if the network is slow. Try again.

CONNECTPRO (PURCHASED OPTION)

- Bar Codes will fail to be read by the Bar Code reader if the HIS/RIS system the bar code was created on has a different language keyboard than the CT system. For example, if your CT system has a French language keyboard then your HIS/RIS must have a French language keyboard. If it is not possible to have the same language keyboard on each system, then manually enter the Accession or Patient ID number or select the desired patient from the Patient Schedule list to display the patient information on the New Patient screen.
- Patients selected from Modality Work List will be displayed with ^ (carets) to define different DICOM fields. The carets are not displayed on the images.

PERFORMED PROCEDURE STEP (PART OF CONNECT PRO OPTION)

- PPS will post the message “Failed to Start” if the remote server is down.
- Screen Save images created in ExamRx or in the Viewer on the Image Works desktop are not PPS aware in this release of software. Also, images created in Reformat, 3D, and Navigator are not PPS aware. For these image types, “INPR” will be posted in the PPS column in the Browser even though PPS is not enabled.

3000 IMAGE SERIES

Reconstruction

- Performance of the recon management screens will degrade as the number or images queued or the recon queue becomes larger. Delays will be seen in transition or closing of retro recon screens and queuing of images for retro recon.
- Be cautious in the number of exams in the recon queue at one time. It is ok to start retro recon on a previous exam and then scan another patient, but care should be taken to not load too many series for retro recon at once.

Display

- Paging and image review with next prior may slow when reviewing large series.

3000 Image Series continued

Archive

- An image series of 3000 images can only fit on one side of a 2.3 gb MOD.
- Auto Store will not save the exam if the media size attached is 1.2 gb.
- If a 1.2 gb media is used for image archive and the image series is greater than 1500 images archive will need to be queued by image. Select a range of images 1500 or less when archiving to 1.2 gb media.
- If greater than 2000 images are queued to save by image you may see the browser disappear and reappear. While this is occurring it will be impossible to access the browser. To avoid this queue less images at a time when saving by image.
- Images may fail to save if saving large groups of images. Verify that all desired images are saved before deleting them from the system disk.

Network

- When queueing a large number of enteries for send or receive, there may be delays when accessing the network queue. Typically this will be seen when 200 or more enteries exist.

Reformat

- Reformat Detail can only load 1000 images. Either select a range of 1000 images or use Reformat standard

3D/Navigator/Dentascan/Add Subtract

- These applications can only load 1000 images. Hilight the desired range of 1000 images by selecting the first image desired, hold the shift key down and hilight the last image desired then select the desired application

RECONSTRUCTION

- If images fail to recon the following should be used in order:

  Select [Recon Management], [Unsuspend Queue]

  If the image still fails to recon

  Select [Recon Management], [Restart Queue]

  If this still fails Shutdown and reboot the system.
- Lung Algorithm
  - Provides edge enhancement between structures with large density differences, such as calcium and air.
  - Enhances the contrast of small objects. For best viewing and film quality, select a window width of 1000 to 1500 and a window level of -500 to -600.
  - Increases CT number values at the edge of high contrast objects. When planning to take CT number measurements of vessels or nodules in the lung, please check and compare your results with Standard algorithm images. (ROI and Histogram functions use CT numbers.)
  - The edge enhancement provided by Lung Algorithm may not be appropriate in some clinical cases. Please take individual viewing preferences into account when you choose Lung algorithm.

Reconstruction continued

- The dynaplan screen may fail to update if Priority Recon is selected during scanning
- Images may show a shading band at the outside edge of a 50 cm DFOV. Change the DFOV to 49.8 to avoid this.
- Retro recon may not be able to get the same image locations as prosepctive recon. This is due to rounding in the start and end location. To avoid this mismatch prescribe start and end locations that are even numbers.
- Retro recon list service exams from Fast Cal. Do not attempt to recon this data, the SRU will shut down. These exams are listed with exams numbers that beign with 50000.
- SmartStep series are listed in Retro Recon but cannot be selected.
- Axial or Helical series will not be listed in retro recon if a scout series is confirmed but not scanned. A reboot will be required to dislay the series on the Retro Recon List Select screen.
- Anatomy scanned off center from ISO center such as extremities, may have loss of resolution when data is acquired helically. Always make sure to center to anatomy of interest as close to ISO center as possible.
- Cine acquisitions may have retries or image suspension. Unsuspend images in recon management.

RETRO RECON

- Two decimal points cannot be entered, i.e. 3.75 for image interval if column edit is used. Edit each scan group (row) if an interval with 2 decimals is required.
- If a 1 rotation 4i or 2i axial scan is prescribed using add group in ViewEdit with a Superior to Inferior scan direction is retro reconned, the resulting images will be reconned Inferior to Superior. The image locations are correct, but the image numbers will not match the prospective images.
- If you delete queued Retro images the image space reserved for those retro recons is not given back. You will not get the space back until you reboot the system.
- Only start Retro Recon when scanning is complete. Do not delete queued retros while scanning is active, scanning could stop.
- If maximum A-P or R-L offset is selected in Retro Recon, one image may fail to reconstruct. Select an offset that is 0.5mm less than the maximum value allowed.
- The start/end locations of a retro reconstruction may be different from Recon 1 and 2 and 3 when a PMR with a thicker slice than Recon 1 is prescribed for the series.
- Retro Recon may hang and core. If the Retro recon screen fails to display or queue retrospective images, retro recon has cored. Reboot the system to recover.
- If an R, L or A, P value is more than half of the Display Field of View in mm. The image annotation will not annotate R and L or A and P on the images. For example if the DFOV is 10cm and the R value is 56mm then the image annotation will show R R. Use a R-L or A-P value which is less than half of the DFOV in mm to avoid this.
- It may be possible to confirm a retro recon prescription with an invalid recon start and end location that results in a recon suspend. To avoid this change image thickness before changing image locations.

RECON MANAGEMENT

- Scout queue entries can be selected in Delete Retro queue entries. Only select Scout queue entries to delete if you are sure you do not need the scout image reconstructed.
- Before saving or restoring scan data make sure the system is idle and no Archive, Network or Filming is active. No other features should be accessed until the save or restore is complete.
- Recon Management may hang while trying to display the menu. Finish the current exam if scanning and shutdown and restart the system to correct the problem.
- If you wish to cancel a Save or Restore scan data, the cancel button will cancel only after the current scan file is saved or restored. A helical scan file contains a large amount of data even though it is one scan file, that file can take up to 30 minutes or more to save. You are unable to scan patients while saving or restoring Scan Data.
  Make sure that you have ample time to complete the save or restore before beginning.
- Scan data save or restore is active when the dialog message indicating saving or restoring scan file is displayed.
- If saving scan data, after the save is complete select Restore Scan Files to verify the data is stored.
- Scan File save is not complete even though the Saved File message has been posted. The problem is due to the time it is taking to unmount the MOD. You may have to wait 5-7 minutes for the MOD to unmount. The write light on the MOD will be green when the MOD is unmounted
- Scan Files restored from a different system will be reconstructed using the product name of the system it is restored on. For example, restore scan files from a H1 SDAS QX/i or a H2 SDAS Plus to a LigtSpeed[16] and the product name on the images will be indicated as LightSpeed[16] instead of LightSpeed QX/I or LightSpeed Plus. All other patient and parameter information is correct.
- Scan save is limited based the size of the MOD. The following lists the maximum scan time that can be saved to a 1.2 or 2.3 gb MOD

speed	trigger rate	max 8 Slice Scan Time 1.2 GB MOD	max 16 Slice Scan Time 1.2 GB MOD	max 8 Slice Scan Time 2.3 GB MOD	max 16 Slice Scan Time 2.3 GB MOD
0.5	1640	27.06309342	13.58856781	49.2056244	24.70648694
0.6	1640	27.06309342	13.58856781	49.2056244	24.70648694
0.7	1400	31.70248086	15.91803658	57.6408743	28.9418847
0.8	1230	36.08412456	18.11809042	65.6074992	32.94198258
0.9	1090	40.71878276	20.4451846	74.03415047	37.17306291
1	984	45.1051557	22.64761302	82.009374	41.17747823
2	984	45.1051557	22.64761302	82.009374	41.17747823
3	984	45.1051557	22.64761302	82.009374	41.17747823
4	984	45.1051557	22.64761302	82.009374	41.17747823

CT PERFUSION2 ON OC – PURCHASED OPTION

- Saving TIFF files of the graph and images is not supported because there is no floppy drive hardware available on the OC.
- The CT Perfusion Control Panel may be displayed on the current desktop if CT Perfusion is launched from the Browser and the desktop is changed before CT Perfusion screens are completely displayed.
- Saving TIFF files of the graph and images is not supported because there is no floppy drive hardware available on the OC.

CT Perfusion2 on OC continued

- Set the color ramp to gray scale for screen saves images if you are going to network the images to another station such as PACS to improve quality of the screen save.
- Initially as CT Perfsuion2 is loading, you may see garbage flash along the left side of the screen.

SMARTSCORE PRO – PURCHASED OPTION

- Only 0.8 and 0.5 second rotation speeds are valid for Prospective Gating.
- A backslash (/) in the Patient ID for SmartScore exams will cause the patient report to fail to print or be stored to floppy.
- If the Confirm button is not available, select the Gating button and turn gating off and click on Accept. Select Gating button, again, and turn gating on and click on Accept. The Confirm button should now be available. Contact your local service representative if you experience this issue for further investigation.

CARDIQ SNAPSHOT – PURCHASED OPTION

- If a lead falls off during acquisition, the heart rate annotation will be inaccurate for the portion of the acquisition after the lead falls off. Images reconstructed for the complete acquisition using Snapshot Segment will initially display correct BPM for the period of time the leads were connected, and then ungated images will be displayed followed by images annotated with abnormally high heart rate where the leads have fallen off.
- If the Gating button is displayed in red and heart rate is seen, this is most likely due to loose connection of the cable at the connection to the EKG monitor or the scanner. Make sure you see a heart rate before you begin the scout scan.
- CardIQ Snapshot Segment annotation is not displayed on the images displayed on the Advantage Windows workstation. The average heart rate and the percent of R-R interval will not be seen. The Cardiac Helical scan type will be annotated as Axial.
- Snapshot Segment images can be networked to a PACS system, however due to an error in the DICOM header, they can be retrieved from the GE PathSpeed PACS system but the image annotation is wrong.

DIRECT3D - OPTION

- If you resize the viewport in Direct3D, the image may not initially be displayed. User the Page Turner in the lower left corner of the viewport to toggle to the time.
- The Direct3D model may not initially build properly. Enter Interactive Review and re-render the model and then the model will build correctly.

SMARTSTEP (PURCHASED OPTION)

- The display process may crash if SmartStep is confirmed from a desktop other than the ExamRX desktop. To avoid this always confirm SmartStep from the ExamRx desktop
- The image display during SmartStep may get out of synch and only display a single image.
- SmartStep screen saves may fail to display in the List Select browser while in SmartStep is active. Once you exit SmartStep the screen saves will be shown in the browser.
- Screen Saved images created during a SmartStep session may fail to show up initially in the List Select Browser. The images will appear in the Browser after exiting SmartStep.

AUTO TRANSFER

- A series may fail to Auto Transfer if [Next Series] then [Create New Series] is selected after a scout scan is prescribed, but not scanned, i.e.; pause and return to view edit is selected.

DISPLAY (APPLIES TO BOTH IMAGE WORKS AND EXAM RX DISPLAY)

- Plus recon mode is not annotated on the Series Text Page. Refer to the image annotation for the Plus recon mode annotation.

DISPLAY

- The Exam Rx List Select and Image Works browsers may in some cases not list an exam, series or image. This may be seen after an interruption of power to the system or the database may be updating slowly. The patient does not need to be rescanned.

  Verify the following:

  - Type the Exam, Series, Image on the accelerator line to display the image,
  - Select another exam in the browser this will cause the browser to update,
  - The browser may update after a slight delay,
  - Reboot the system,
  - A second reboot may be needed if a power fail has occurred.
- The browser may update slowly to reflect Screen Save images created in an exam. If the Screen Save images is not listed in the browser wait for them to appear. The time to appear may be up to 15 minutes if the Automatic Database check and recovery is in progress when the Screen Save images are created.
- In general, wait for a display action to complete before entering in another command.
- With the system configured in French or German language, if a comma is entered in explicit magnify, i.e. 2,3 the comma will not be interpreted as a decimal. The magnification factor is applied without the decimal. In the example shown, 2,3 the image would only be magnified 2X not 2.3X. Always enter a period to designate a decimal and have the correct magnification applied.
- Cross Reference lines will post incorrectly if sort by image location is selected. Make sure that sort by number is selected before posting Cross Reference lines.

Display continued

- Cross Reference lines may not post with some combinations of DFOV and RAS center. RAS centers at the edge of the DFOV and small DFOV's will exhibit the problem. Retro reconstruct the images to a larger DFOV if posting of the cutlines is desired.
- Image annotation may overlap or not be fully shown if a 32 character patient name is entered for 3D, Navigator, Reformat, MID formats, DentaScan images.
- Text and Series pages only display 24 characters for Patient Name.
- Text pages will not show foreign characters entered using the ALT GR key.
- ROI and STD deviation numbers may be reported as zero after a Zoom is applied to an image with a ROI posted. This is due to partial pixel contained in the ROI. Adjust the size of the ROI to include full pixels and recalculate the ROI.
- No text will be entered in the accelerator line if the mouse cursor is over the film composer.
- Some combinations or E/S/I will fail to display an image, if cap locks are on. To select an exam series and image from the accelerator line use the following format when cap locks is on: For example to select Exam 576 Series 2 image 3 type E576 2 3 on the accelerator line. If the desired exam, series or image does not display, then select the desired images from the List Select on the Browser.
- Image location seen on the scanner may differ from that seen on an AW 3.1 system for tilted images. LightSpeed reports the image location as iso-center of the image. AW 3.1 reports image location as the image center for the image. With tilted images there will be a difference in the location numbers proportional to the off center distance times the sin of the tilt angle. The image location can be reported as image center on LightSpeed by using the Report Cursor function.

EXAM RX DISPLAY

- Display may reset it's self after a software problem. You may see display stop or not accept your mouse input. Wait for a few seconds and display will automatically reset. You will have to redisplay the images you wish to work on.
- The MIROI pop up may display but will not function. This will occur if an Auto View or Cross reference viewport is in primary focus (blue border) MIROI is not valid in these viewport types. Place a Free or Auto Link viewport in primary focus and reselect the MIROI button to continue.
- Annotation in the AutoView viewport may appear black and shadowed. Move your cursor over the viewport to reset the annotation to white.
- Trackball paging will not start if paging has been started, (i.e. middle button has been doubled clicked) and no images have been reviewed, (i.e. the trackball was not used) and paging was cancelled. The series must be redisplayed from the browser for paging to start.
- Any Report Pixel or MIROI chart that is not screen saved, but is filmed will show the Exam Series and Image as 1000/1/1 on the film composer icon if E/S/I is selected. This is the exam/series/image number the system assigns to this chart display The patients exam number is listed correctly on the filmed image.
- Image selection from the Accelerator Line will not function if the primary viewport contains a MIROI plot or Report Pixel chart. Use List Select to display a new image.
- Any accelerator commands entered for Series Binding, Annotation levels for display or filming will be saved as defaults

ExamRx Display continued

- ExamRx Display may hang. In some cases switching to the Image Works desktop and back to Exam Rx display may clear the problem. If this doesn't work then perform a system shutdown and reboot the system.
- With paging active (P in the lower left corner of the image) the trackball may adjust window width and window level in other viewports, verify the image has the correct window level before filming the image.
- The right mouse may fail to roam even when the state is set to roam. A system shutdown may be required to clear this problem.
- Display may crash when selecting a 3D object from the list select browser. When a 3D object is selected, 3D should be selected from the browser.
- Images may not be displayed after a switch between Auto View or Image Review layouts. This is due to a restart in the display process. Redisplay the desired images in the viewport to continue.
- The first time the trackball is used for paging after a system reboot window level will be changed instead of paging the images even though the P icon is displayed. Click once to switch to W mode then click again to place the trackball in paging mode.
- The AutoView or Auto Film viewport may display a very magnified image. This image may be filmed in Auto Film. Review your films and verify that all filmed images look correct.
- The mouse cursor may become locked in either the roam or zoom mode in the Auto Link viewport and will be oppsite of the selection for right mouse control in Routine Display. Try switching layouts first, if this does not revert the mouse to the correct mode then reboot the system.
- Viewport format may not be able to be changed and requires switching layouts to correct. This occurs if F3 or Shift F3 is used for filming and next image is used. The viewport format is now locked. You will have to switch to another review or Auto View layout to correct.
- Drag and drop for a scout with a large number of cross reference lines may take longer to capture. Click left and hold the mouse on the scout with cross reference lines for a second before dragging to the film composer.

IMAGE WORKS DISPLAY

- WW and WL adjustments will only be maintained on individually selected images if the middle mouse button is used. If the accelerator line or Presets are used all images will update regardless of primary or secondary focus of the images.
- Any accelerator commands entered for Series Binding, Annotation levels for display or filming will be applied and not held if the viewer is closed. Use User Preferences in the viewer to save the settings as a default if desired.
- Window/Level does not display stored value set in the Viewer when +/- series is used. The W/L used is the same as the last series displayed.
- Images created in Add/Sub are displayed with value of WW 4098 and WL 1024.
- If paging is selected while the system is in Compare Mode, the upper left viewport is left blank. Return to the Browser and select the desired series for paging and restart the Viewer.
- Accelerator Command Line entries for E/S/I do not act properly. The first image in the series is always displayed instead of the requested image within the series.
- The F10 function key does not work for preset window/level for Reformat, Navigator, and 3D Analysis.
- Cross Reference line numbering along the top of a sagittal scout may be listed in offset manor.
- The new features in User Preferences on the Image Works desktop are not translated.

Image Works Display

- The Square Viewport setting is not remembered between system reboots and will always default to off at reboot.
- RAS measurements may be off on screen saved images if the viewer format used was 1 on 1. To avoid this use a 4 on 1 viewport to screen save images that contain measurements.

VARIVIEWER

- If you resize the viewport in Direct3D, the image may not initially be displayed. User the Page Turner in the lower left corner of the viewport to toggle to the time.
- It is possible to get two different DFOV for a combined VariViewer session if Column edit is used. Use Row edit to avoid this.
- If the scout is taken after the series used for VariViewer operation, the system may be unable to find the scout for graphic batch prescription. Manually enter the range for the batch prescription.
- Any zoom applied to a Variviewer image is not maintained as you page through the images.
- VariViewer session may not be able to be started if there multiple Pause scans during the acquisition of data.
- VariViewer is not compatible with a gantry tilt.
- Switching back from a 768 viewport is VariViewer causes the VariViewer viewport to to be forward instead of the Auto View viewport. Use the page turner to toggle back to the aut view Viewport.

ADD SUBTRACT

- Comb images will not contain the e/s/i used for the added image.

EDIT PATIENT DATA

- The system may hang if Edit Patient is started while the following operations are in progress:

  Network Receive

  Prospective or Retrospective reconstruction is active

  Archive Restore

  To avoid this, confirm Edit Patient during idle times in network receive, active recon or archive restore.
- Patient age may be seen as zero in Edit Patient Data even though a birth date is entered. Enter the birth date again and patient age will be calculated.
- If duplicate accession numbers exist, Edit Patient Data will not allow the exam to be modified.
- If there is a space at beginning or end of the hospital, Edit Patient Data will fail. Contact your field engineer to update the hospital name correctly in the system reconfiguration.

REFORMAT

- When selecting reformat wait until the % loading dialog is displayed before switching to the Exam RX desktop. This will avoid having to select Load slices to start the reformat model build when returning back to the Image Works desktop.
- The reformat location will not be annotated on images filmed when Printer is selected in Batch Mode. If it is desired to have the reformat location annotated, then use Save in Batch Mode and film the images through Print Series.
- In Batch mode, images reconstructed using “Rotation” or “Oblique” modes will start with number 2 while images created using “First View/Last View” start with 1. In Rotation and Oblique modes, a plan of the reconstructions is screen saved and becomes image 1 for the series.
- Occasionally, the color for the viewports and control panel will change because there is not enough colors available in the resource file to support reformat. To recover, quit reformat and restart the application.
- Window/Level for images are not displayed using the value stored in the image. Use the window/level function keys or selects within Reformat to reset the W/L.
- Reformat will only allow filming in Batch to the first 30 cameras in the list. If another camera is desired save the batch images to disk and use Print Series to film the images.
- Images saved from batch mode may appear to have jagged edges or not have optimal Image Quality. Manually save the images to improve image quality.
- Large image series may fail to load when using Reformat detail. This can be caused by small spacing interval between images or image sets greater than 1000 images. Select a sub set of images if this occurs or use Reformat Standard to build the model.
- A series that has more than 1 group with 0.625mm slice thickness will fail to load for reformat or will fail in Volume rendering on the AW. This is due to a mismatch in the spacing between the groups. To avoid this have only one scan group if possible, otherwise selectively hilight the images from the first group for Volume Analysis.
- Reformat has trouble loading images with different parameters. Normally reformat will only load images that have similar parameters even if the series contains groups with different parameters. This will not occur if an image from the second group was selected. You must select the range of images desired from the second group.

FILMING

- Images or Auto Film control buttons may fail to display in the AutoFilm viewport. This may occur after a switch between desktops. To display the images change to a different AutoView layout, then switch back to your desired layout.
- Don’t let the auto film viewport back up with images to be filmed. Start Auto Film as soon as possible to keep caught up on filming.
- Any of the Auto Film control buttons may be activated when the cursor is over the button and the space bar is selected. Place the cursor in the Auto Film viewport only when needed to avoid this problem.
- Auto Film may fail to film. This may occur when the Auto Film viewport selection shows active and the Auto Film status shows paused. Toggle the [Pause] button in the Auto Film viewport and select [Start New Sheet] or [Continue Same Sheet] to restart filming.

Filming continued

- When scanning multiple one rotation axial groups, images are filmed out of order when Auto Film is selected. Turn Auto Film off for these scan groups.
- Intermittently, Auto Film will automatically print the last page when Add Group is selected.
- Occasionally, the last image will not print. Cancel Auto Film and manually film the image to the Auto Film Composer. If scouts and reference lines where prescribed, they will need to be filmed manually
- Auto Film may fail to display images if recon is having trouble reconstructing the image or if images have failed to install in the database. This may also occur if a large exam has not reconstructed all images and scanning is started on a new exam. The following dialog will be posted if Auto Film cannot display images:
  - Exam 100
  - Series 2
  - Cannot find 10 images
  - Skip Missing images Continue Cancel Film Series

  You are are presented with 3 choices:

  1. Skip Missing images will skip the images that Auto Film cannot find. For example if Auto Film had filmed images 1-10 and you then got the message that 10 images were missing and selected Skip Missing Images, Auto Film would then start filming again with image 21.
  2. Continue will look for the images again, if they are not found then the dialog will be posted again. Before selecting continue verify that recon is active, that the images have been reconstructed, the missing images are not suspended or paused in the Recon Queue and the images are able to be displayed. If the images are suspended or paused in the recon queue, then unsuspend or restart recon to reconstruct the images before selecting Continue.
  3. Cancel Film Series will cancel Auto Film for the series currently being Auto Filmed. The images will then need to be manually filmed.

  If you get this dialog because you have begun an exam while the previous exam still has images to reconstruct, then Select Continue and select Pause Auto Film, resume Auto Film when the Exam Series and Images from the previous exam have reconstructed.

- When prescribing Mag factor for auto film, scout images if the scout is longer than 500 mm then use a magnification factor less than 1 to display the entire scout.
- The manual film composer may display when confirm is selected for scanning if the composer had been closed by selecting iconify in the upper right corner of the composer.
- It's best to run Print Series from only one desktop at a time.
- It's best to resolve any paused queue entry as soon as possible.
- Print Series may pause automatically under heavy system load.
- If Anonymous Patient is selected for an exam where Auto Film is in progress, the system may fail to cross-reference the slices on the scout. If Anonymous Patient is required for the series, wait till Auto Film is complete for the series before proceeding with Anonymous Patient.

Filming continued

- To install a SMPTE pattern select the followin . Once installed the images will be available for viewing from the Image Works browser or List Select in Exam RX. And will be listed as Exam 1000. The patient Name will be listed as SMPTE.

  Select Service Desktop

  Select [Image Quality]

  Select [Install SMPTE Image]
- Full Annotation instead of partial annotation will be filmed when using F3 (Film MID) in Image Works desktop when full annotation is selected.
- The format built in a protocol may be changed when the protocol is used with a message that the format was changed due to an invalid format, even though the format is valid. Verify in Auto Film set up is the format you desire before confirming scan.
- For Multi-Image Display (MID) in AutoFilm, filters, gray scale and orientation selections are applied only to the image in the upper left viewport. If there are an odd number of images in the series, the filters, gray scale and image orientation is applied to the first group only.
- Auto Film may stop and the Continue Same Sheet button will be displayed. The Continue Same Sheet button will be active, but when selected will not restart filming. Use Manual Film to complete filming the exam and reboot the system before starting the next exam
- If the F1 key is used to film the Text Page ROI, Exam Text Page and Series Text Page, the text pages will be filmed to the film composer last selected in the pop-up for text page. Use the filming selections in the text page screens to film the information displayed.
- The message “Film formatting in progress. Please retry.” May be seen when filming to the Manual Film composer when the system is busy with other simultaneous operations such as recon, network, auto film and scanning. Retry manual filming when the system is less busy. A system reboot may be required to recover.

ARCHIVE

- The feature status area or the browser may report an Exam has saved even though all images have not been saved to MOD. Scroll through the images in the image window. Hilight the images listed as Archive N and resave them.
- Restoring Exam, Series or images that already exist on the system disk will not post a message that the images are restored or that they already exist. If you have restored images, but get no message that they have been restored, verify that they don’t already exist on the system disk.
- When the archive media gets close to being full, the system will always look to see if the exam, series or image can fit on the MOD, if it is desired to have exams sequentially on a MOD, place a new MOD in the drive when first notified the disk may be full.
- If Label is selected in Archive, finish the Label process; don’t select cancel on the Label pop-up. Archive will be non functional, a system reboot will be necessary to recover.
- It’s best to resolve any paused queue entry as soon as possible.
- To minimize corruption of MODs, it is extremely important the MOD media be detached and removed from the MOD drive before doing a shutdown or recycling power to the system.
- When restoring from a MOD, it is recommended to write protect the media before placing it in the drive. If the system is unable to read the media, remove the write protect and see if the system is able to perform a recovery of the media to access the data.

Archive continued

- If the MOD can not be detached or will not eject from the drive, do a shutdown and eject the MOD once the message "Hit any key to start the system" is seen. Do not use the screw to eject the disk.
- If you get a timeout message when trying to access the MOD, the MOD drive may no longer recognized by the system. Perform a shutdown to re-establish communication between the system and the MOD drive.
- Even though the Sony MOD drive states that it is a 5.3 gb drive. The archive software only supports a maximum size of 2.3 gb. 2.3 gb 512 bytes per sector is the preferred archive media size to utilize on the system.

NETWORK

- LightSpeed images will **not** transfer if Advantage Net protocols is used. Always use DICOM protocol to send LightSpeed images.
- LightSpeed images **cannot** be sent to a HiLight Advantage, HiSpeed Advantage or CT Independent console. These systems do not support DICOM receive.
- Some 3rd party workstations may fail to receive LightSpeed scout images. This is due to the matrix size of the scout image. These stations do not support receiving matrix sizes greater than 512. Some LightSpeed scouts images have matrix sizes greater than 512. If it is desired to have the scout image on the workstation, Screen Save the scout and then transfer the screen save image to the workstation.
- If you query a LightSpeed system from a CT/I or Advantage Windows system in Advantage Net, only Advantage format image exams will be displayed, No LightSpeed exams will be shown. Always query a LightSpeed system using DICOM protocol.
- Lateral scouts displayed on an AW 3.1 system will initially display with zero rotation; they should display with a 270-degree rotation. Rotate the image in the left direction using the rl command on the command line to display the scout in the desired format.
- Images networked to a Advantage Windows 3.1 workstation running software versions prior to 3.1_07 will display a DFOV less than what is displayed on the scanner. This is due to the AW not taking in account the pixels under the focus border of the viewport.
- Images networked to a AW 3.1 will have the following annotation missing:
    - Accession number,
    - Date of Birth,
    - 4i, 2i or 1i recon mode on Axial, Retro Axial or Axial Reformatted images,
    - Series type on Retro images,
    - Table Speed on Helical images,
    - Images are annotated +C when Oral Contrast is used. View the Series Text Page to see if IV contrast was used.
    - CardIQ annotation for BPM and percent of R-R interval and scan type will displayed as Axial.

Network continued

- The number of images indicated in a series may be incorrect on the remote browser when a query is made from an AW or CT/I station.
- Series types may be listed differently when images are networked to an Advantage Windows 1.2, 2.0 or 3.1 system than that shown on the LightSpeed system
- Exams with Swedish, German or French characters in any Patient Info field will not be transferred to an Advantage Windows.
- When using DICOM protocol, the entered host name must match in spelling and case sensitive, otherwise a connection error message will be displayed.
- It's best to resolve any paused queue entry as soon as possible.
- Images that have been edited using Edit Patient networked to a RadStore system will fail to archive to the RadStore DVD. The images must be archived on the CT system MOD.

IMAGE MANAGEMENT

- Don't let image space fall below 200 images on a single disk system (130,000-image storage). This will ensure there is room on the system disk to confirm scans and to install reconstructed images.
- Removal of images should be done with the system in an idle state. Failing to remove images at idle times can cause image install timeouts that can then cause recon to shutdown and can cause the browser to fail to respond. The user may also see a blank list in the browser. The system should not be shutdown until the image space has been updated fully from the remove. The length of time to synchronize the database is proportional to the number of images being removed. This will assure that all images are reconstructed, displayed and installed into the database without error and that image space is update correctly.
- Exams that contain SmartStep may fail to remove. Reboot the system to remove the lock on the SmartStep series.

MOBILE

- Mobile systems scanning using the back-up power generator instead of shore power will experience vibrations as the generator compensates for sudden electrical loading as the X-Ray turns on. This is normal operation for generator verses shore power. To ensure these vibrations experienced on generator power do not affect image quality, the anatomy being scanned should be secured to the cradle using the restraint straps.
- The CT Mobile software allows the user to pre-program different site names and networking addresses for sites visited by the mobile. If the system configuration is moved to an external subnet instead of the internal subnet, the network for the camera and AW will need to be redefined for the external subnet.

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GE Medical Systems

P.O. BOX 414
MILWAUKEE WISCONSIN 53201-0414 U.S.A.